



VITEK FITNESS

NUTRITION ESSENTIALS GUIDE

This guide focuses on practical nutrition principles that support performance, health, and long-term consistency.

MACRONUTRIENTS

PROTEIN

- Proteins are made of amino acids. Your body needs all 9 essential amino acids (EAAs) to recover and build muscle.
- High-quality proteins (complete amino acid profile): animal protein - egg, chicken, fish, dairy - yogurt, quark, whey.
- Plant proteins should be combined to form a complete amino acid profile.
- **1 gram of protein = 4 kcal**

- If eating plant protein: combine **legumes + grains** in a 1:2 ratio (e.g. lentils + rice, hummus + whole grain bread).
- Aim to consume enough protein throughout the day - especially around training.
- If you eat enough protein 1 - 2 hours before training, your body can use it during and after for muscle recovery. You don't necessarily need protein immediately after training if you had enough before.
- Recommended : 1,8 - 2 g of protein per kilogram of lean body mass (active body weight).

CARBOHYDRATES

- Carbs provide glucose, your body's main energy source.
- **Complex carbs** (e.g. oats, brown rice) provide long-lasting energy.
- **Simple carbs** (e.g. fruit, honey) are ideal before/after training for quick energy.
- **Fiber** aids digestion and keeps you full. Aim for 25–35g/day.
- **1 gram of carbohydrate = 4 kcal**

- Fruit contains sugar - fructose is natural but still sugar. **Be mindful of overall sugar intake.**
- Avoid artificial sweeteners when possible - they may negatively impact your gut microbiome.
- **Combining simple sugars (like fruit) with protein, fat, or complex carbs slows digestion.** This prevents a rapid insulin spike and makes it a healthier option compared to eating sugar alone.

FATS

- Fats are essential for hormone health, brain function, and nutrient absorption.
- **Healthy fats:** olive oil, nuts, seeds, avocado, fatty fish.
- Limit unhealthy fats: trans fats, highly processed meats.
- **1 gram of fat = 9 kcal**

OIL

- **Extra virgin olive oil** is a great choice for everyday kitchen use due to its antioxidants and balanced omega-3 to omega-6 ratio. However, its smoke point is lower (~160–190 °C), so frying should always be done on lower heat to avoid oxidation.
- **Ghee and coconut oil are very heat-stable** and suitable for frying, as they oxidize less easily, though both are high in saturated fat and should be used in moderation.
- **Seed oils** (such as sunflower, soybean, or corn oil) **are less recommended.** They are typically high in omega-6 fatty acids, which can contribute to inflammation when consumed in excess, and they are more prone to oxidation when heated.
- The **smoke point** is the temperature where oil starts to burn and oxidize, producing harmful and potentially carcinogenic compounds.
- **One tablespoon of oil = approx. 120 kcal**

MICRONUTRIENTS

VITAMINS & MINERALS

A balanced diet provides the foundation for meeting most micronutrient needs.

However, due to modern lifestyle factors, limited sun exposure, dietary patterns, and variability related to food choices, processing, and storage, certain nutrients - such as **vitamin D, omega-3 fatty acids, and magnesium** - are commonly insufficient and may benefit from targeted supplementation.

- **Magnesium** is important for muscle relaxation, energy production, and sleep. Good sources: pumpkin seeds, nuts, dark chocolate, leafy greens.
- **Vitamin D** supports immune function, bone health, and mood.
 - Due to limited UVB exposure at northern latitudes (e.g. Berlin), vitamin D levels are often low throughout the year.
- **Omega-3 fatty acids** support heart, brain, and joint health, but intake is often low because many people do not consume fatty fish regularly. Supplementation can help ensure adequate EPA and DHA intake.

SUPPLEMENTS

TIMMING & ABSORPTION

VITAMINS & MINERALS

- **Vitamin D** is best taken in an oil-based form and together with vitamin K2
 - A commonly used guideline is **50–100 µg of vitamin K2 (MK-7) per 1,000 IU of vitamin D3**.
 - A low daily dose (around 800–1000 IU) can be considered year-round, unless regular sun exposure without heavy sunscreen use is ensured.
 - High doses of vitamin D (e.g. 20,000 IU or more) are usually used short term to correct deficiencies and should ideally be based on blood testing.
 - When higher doses are used, adequate vitamin K2 intake becomes more important.
- **Omega-3** supplements are best taken with meals to improve absorption and reduce digestive discomfort.
- **Iron** supplements should not be taken with coffee, tea, or calcium. Vitamin C improves iron absorption.
- **Magnesium** - the timing depends on its form:
 - **Magnesium glycinate/bisglycinate** is often preferred in the evening due to its calming effect.
 - **Magnesium citrate or malate** can be taken during the day and may support daily activity and energy metabolism.
- Coffee mainly interferes with mineral absorption (especially iron), not vitamins or fats.
- **Electrolyte / isotonic drinks** help replace minerals lost in sweat during intense training.

PROTEIN POWDER

Protein powders are a practical way to increase daily protein intake when food alone is not sufficient. They are supplements, not replacements for whole foods.

Total daily intake matters more than timing or supplement type.

WHEY PROTEIN

- **Whey isolate** is usually best tolerated and contains very little lactose and fat.
- **Whey concentrate** is also effective but may be less suitable for people sensitive to lactose

PLANT-BASED PROTEIN POWDERS

Plant-based protein powders can be effective, but protein quality and amino acid balance require more attention.

Single plant proteins often lack one or more essential amino acids, while **blended proteins improve amino acid completeness** and overall effectiveness for muscle growth and recovery.

What to look for:

- **Protein blends** rather than single sources
(e.g. pea + rice, pea + hemp).
- Adequate leucine content or higher total protein per serving.

CREATINE

CREATINE MONOHYDRATE is one of the most researched and effective performance supplements. It supports strength, power, muscle mass, and recovery

- Timing (pre vs post workout) is not critical - consistency matters more
- Drink with water and ensure adequate daily hydration.

Daily intake

- **3 g/day** for lighter individuals ($\approx$ under 70 kg) - general training.
- **5 g/day** for heavier individuals ($\approx$ 70–90+ kg) or those with higher muscle mass - regular strength training or high training volume.

Loading phase (optional)

- 20 g per day split into 4 doses for 5 - 7 days.
- After loading, continue with **3 - 5 g daily**.
- Loading is optional - long-term intake yields the same benefits.

Creapure® creatine monohydrate is a branded form known for high purity and quality control. You don't *need* Creapure® to get benefit - generic creatine monohydrate works well too - but Creapure® offers a quality assurance advantage.

FAT LOSS

- Fat loss requires a calorie deficit but without starvation.
- Focus on nutrient-rich, filling foods rather than simply eating less of everything.
- Avoid sugary drinks, fried foods, snacks, and alcohol.
- Be aware that sauces and especially oils add many calories; one tablespoon of oil has about 120 kcal

Important: Never eat below your Basal Metabolic Rate (BMR). Going under this level slows down your metabolism and makes fat loss harder in the long run. Always aim for a sustainable deficit above your BMR.

FAT LOSS & TRAINING

- **Building muscle** supports fat loss, as muscle tissue increases daily energy expenditure and helps maintain a higher metabolic rate over time.
- For **sustainable fat loss**, a combination of strength training, moderate cardio, and daily movement is often more effective than relying on cardio alone.

Cardio training can burn a higher number of calories during the session due to an elevated heart rate. However, very intense cardio may lead to increased fatigue, which can reduce overall daily movement and activity levels afterward.
- As a result, total daily calorie expenditure may not always be higher despite a hard cardio session.

MUSCLE GROWTH

- Muscle growth requires **enough energy and protein** to recover and adapt.
- **A small calorie surplus** supports muscle gain, while eating much more than needed mainly increases fat gain.
- **Consistency** in daily intake **matters** more than perfect meal timing.

MUSCLE GROWTH & TRAINING

- **Resistance training** provides the primary stimulus for muscle growth by challenging the muscles and signaling adaptation.
- **Progressive overload** over time (e.g. increased resistance, volume, or intensity) is necessary to continue building muscle.
- **Cardio training** has health benefits but contributes little to muscle growth compared to resistance training and in a high volume it **may interfere with recovery if excessive**.
- For effective muscle growth, prioritize structured strength training, adequate recovery, and sufficient calorie and protein intake.

Sleep is essential for physical recovery, hormonal balance, and overall health.

Insufficient sleep negatively affects muscle recovery, performance, appetite regulation, and decision-making around food.

Poor sleep can increase hunger and cravings while reducing training quality and recovery capacity.

BUILDING SUSTAINABLE HABITS

- Eat whole foods, stay consistent, and hydrate well.
- **Track progress by how you feel:** energy, sleep, strength, mood.
- **Nutrition is not about perfection but building sustainable habits.**
- If you don't have a meal plan from me, try to find easy meals you can batch cook in multiple portions.

Planning your day around meals can be helpful, but flexibility is important. It takes time to find a routine that works, and it doesn't need to be perfect. Occasional missed meals or changes are normal. **The purpose of planning is to stay mindful of your eating habits and support consistency, not to create pressure.**

Skipping a planned meal occasionally for social reasons is okay; just aim for enough protein and healthy choices when you do eat.

STORING FOOD

- If cooking for multiple days, **portion food into separate containers** to cool faster, making it easy to reheat only what you need.
- Store food airtight in the fridge and **consume within 2 - 4 days** or freeze for longer storage.
- Reheat food thoroughly and **avoid reheating the same portion** multiple times

- **Adding lemon juice** can help keep food fresh longer by slowing oxidation and bacterial growth. This can be helpful for cut fruits and vegetables, and when used in cooked foods such as rice or chicken that are stored in the refrigerator.

MEAL TIMING

- For supporting muscle growth and recovery, **spreading protein intake across the day** (for example 3 main meals + 2 protein-containing snacks) can help maintain more stable protein availability and support muscle protein synthesis. *This approach is beneficial but not mandatory. If a different eating pattern fits better into your daily routine, consistency matters more than meal frequency.*
- **After training:** eat protein + carbs to support recovery in case you haven't eaten before the training.
- **In the evening,** avoid very large meals close to bedtime. Aim to finish eating around 2 hours before sleep to support digestion and overnight recovery. A high-protein dinner helps provide amino acids for muscle repair during sleep.

Fasting or time-restricted eating can be one possible eating structure

- A daily fasting window of 10–12 hours already supports basic cellular recovery processes and is often naturally achieved overnight.
- Longer fasting periods (e.g. 16 hours) may be harder to combine with adequate calorie and protein intake, especially for physically active individuals aiming for muscle growth or recovery.
- Training in a fasted state can be tolerated by some people if sufficient food is consumed afterward, but this is highly individual and not suitable for everyone.
- Choose fasting only if it fits your lifestyle, supports training performance, and allows you to meet your nutritional needs consistently.

PLANT BASED DIETS

micronutrients & supplementation

People following vegetarian or vegan diets may benefit from a broader focus on supplementation to ensure adequate micronutrient intake.

- **Vitamin B12**
 - Naturally found almost exclusively in animal products. Fortified foods can help but are usually not sufficient on their own.
- **Calcium**
 - Calcium from many plant foods is poorly absorbed.
 - Calcium-set or calcium-fortified tofu is one of the most reliable plant-based calcium sources, with good absorption.
- **Iron**
 - Iron from plant foods is absorbed less efficiently than iron from animal foods.
 - Food preparation methods such as soaking, sprouting, and fermentation can further enhance absorption.
- **Zinc**
 - Plant foods contain zinc, but it is less well absorbed.
 - Supplementation is recommended especially in physically active individuals.
- **Iodine**
 - Intake can be low if iodized salt or sea vegetables are not used.
- **Omega-3 fatty acids**
 - Plant sources (flaxseed, chia seeds, walnuts) provide ALA.
 - Conversion of ALA to EPA and DHA is limited.
 - Algae-based omega-3 supplements are recommended to ensure adequate EPA/DHA intake.